TENSORING A QUASI-ISOMORPHISM OF K-FLAT COMPLEXES AGAINST ANY COMPLEX IS A QUASI-ISOMORPHISM

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Allowed standard prerequisites (editorial).

- Existence of a K-flat resolution of any complex. Lemma \ref{lemma-K-flat-resolution}, cited at proof lines 14472-14473. Supplies the K-flat complex replacing the arbitrary complex.
- Tensoring a K-flat complex against a quasi-isomorphism of K-flat complexes is a quasi-isomorphism. Lemma \ref{lemma-K-flat-quasi-isomorphism}, cited at proof lines 14485-14486. Gives the vertical and top horizontal quasi-isomorphisms of the square.

Source reference. Original lemma 064L: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7870–7870. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 06Y4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8873–8873. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 06Y0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8869–8869. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let R be a ring. Let $\alpha : P^\bullet \to Q^\bullet$ be a quasi-isomorphism of K-flat complexes of R-modules. For every complex $L^\bullet$ of R-modules the induced map*

$$Tot(id_L \otimes \alpha) : Tot(L^\bullet \otimes_R P^\bullet) \longrightarrow Tot(L^\bullet \otimes_R Q^\bullet)$$

is a quasi-isomorphism.

Proof. Choose a quasi-isomorphism $K^\bullet \to L^\bullet$ with $K^\bullet$ a K-flat complex, see Lemma 06Y4. Consider the commutative diagram

$$\begin{array}{ccc} \mathrm{Tot}(K^\bullet \otimes_R P^\bullet) & \longrightarrow & \mathrm{Tot}(K^\bullet \otimes_R Q^\bullet) \\ \downarrow & & \downarrow \\ \mathrm{Tot}(L^\bullet \otimes_R P^\bullet) & \longrightarrow & \mathrm{Tot}(L^\bullet \otimes_R Q^\bullet) \end{array}$$

The result follows as by Lemma 06Y0 the vertical arrows and the top horizontal arrow are quasi-isomorphisms. □